
Robust Blockchain-based Federated Learning

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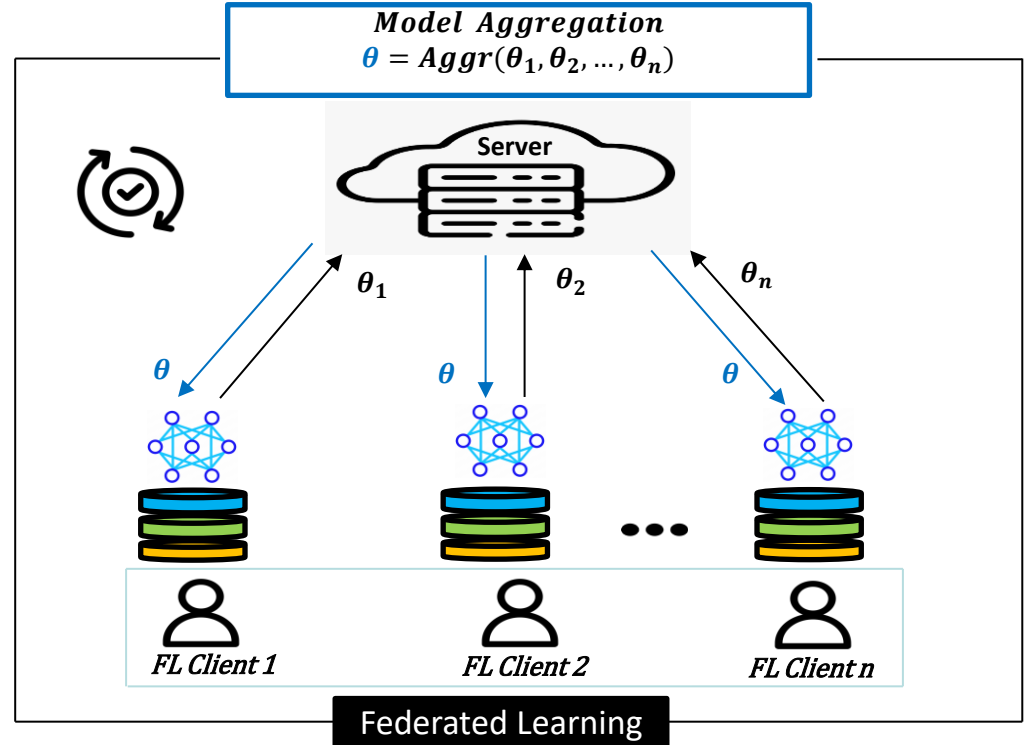
20/02/2025

ICISSP Conference

Federated Learning

- **Federated Learning (FL)**

- Collaborative learning
- Each client locally trains
- Global model = Average of local models
- Iterate until optimal accuracy

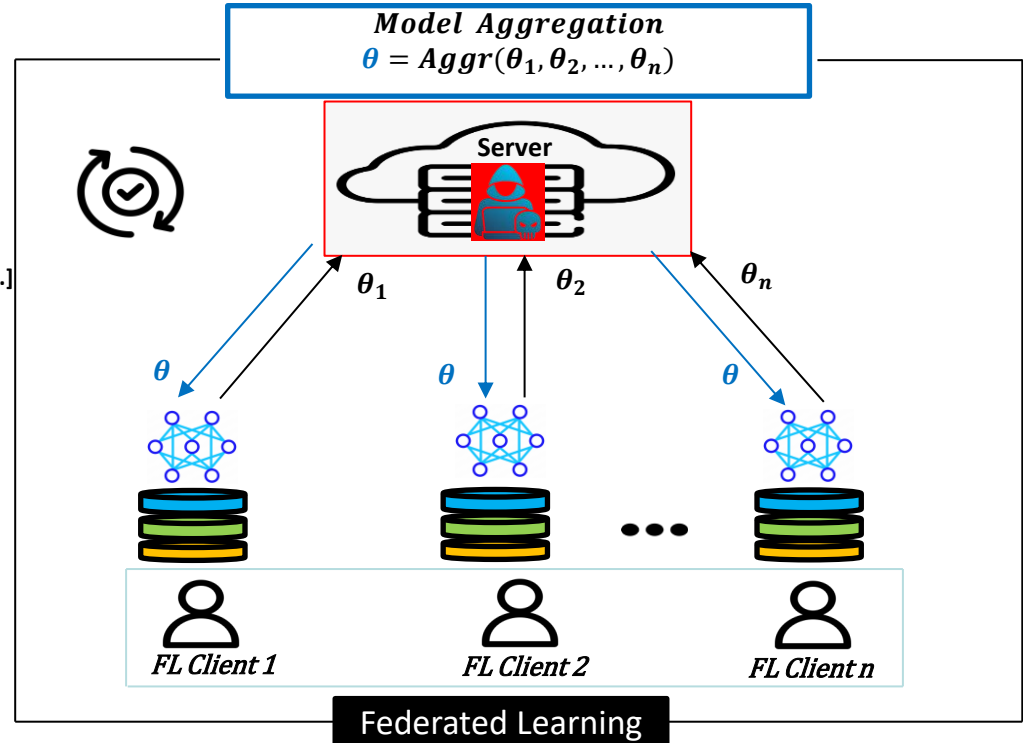


Outline

- Privacy & security challenges in FL
 - Privacy attacks
 - Aggregation forgery
 - Byzantine attacks
- Our solution
 - Blockchain for integrity
 - Trusted Execution Environment (TEE) for privacy
 - Smart contract for Byzantine-robustness
- Experimental evaluation
 - Model accuracy
 - Overhead cost

Privacy attacks

- **Membership inference attack**^[Shokri et al.]
 - Infer data point in dataset
- **Data property inference attack**^[Ganju et al.]
 - Infer sensitive attributes in dataset

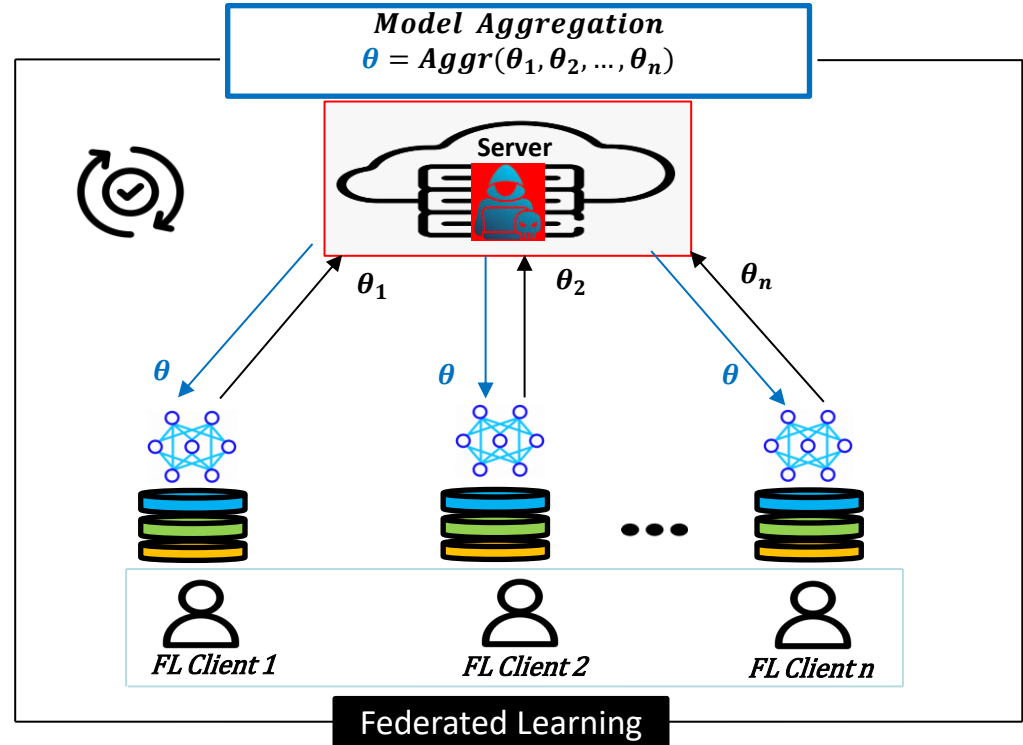


R. Shokri, M. Stronati, C. Song, V. Shmatikov, Membership inference attacks against machine learning models, in: 2017 IEEE Symposium on Security and Privacy, 2017, pp. 3–18.
K. Ganju, Q. Wang, W. Yang, C. A. Gunter, N. Borisov, Property inference attacks on fully connected neural networks using permutation invariant representations, in: 2018 ACM SIGSAC Conference on Computer and Communications Security, 2018, pp. 619–633.

Aggregation forgery

- **Global model degradation** [Buyukates et al.]

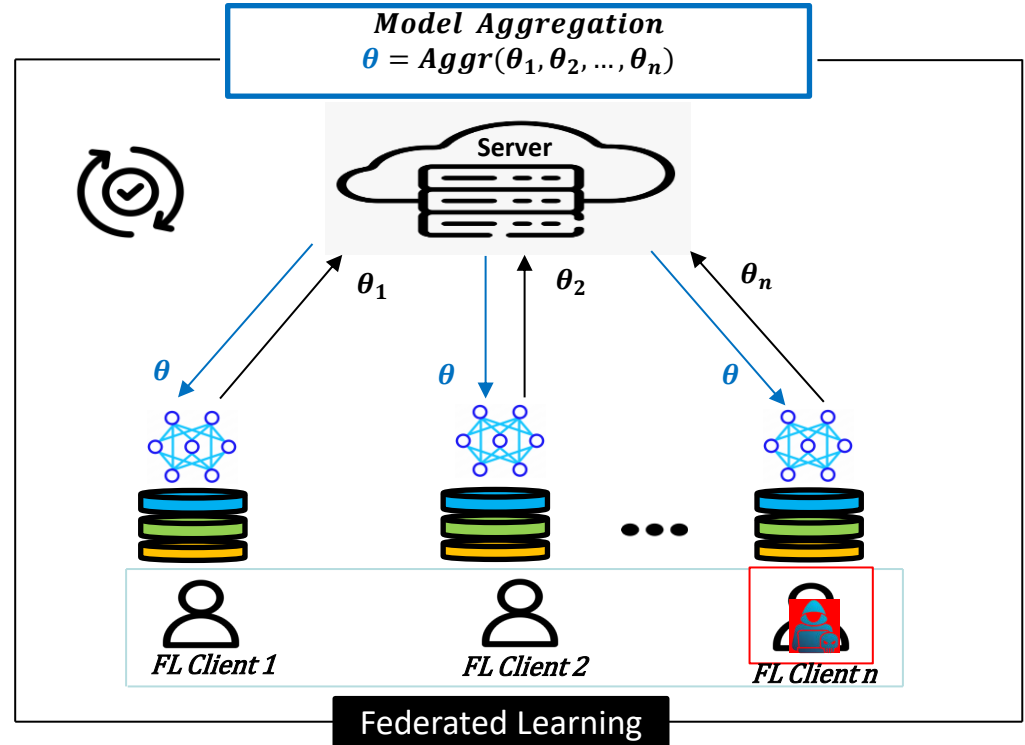
- Incorrect predictions
- Biased outputs



Buyukates, B., So, J., Mahdaviyar, H., and Avestimehr, S. (2022). Light verifi: Lightweight and verifiable secure federated learning. In Workshop on Federated Learning: Recent Advances and New Challenges (in Con-junction with NeurIPS 2022)

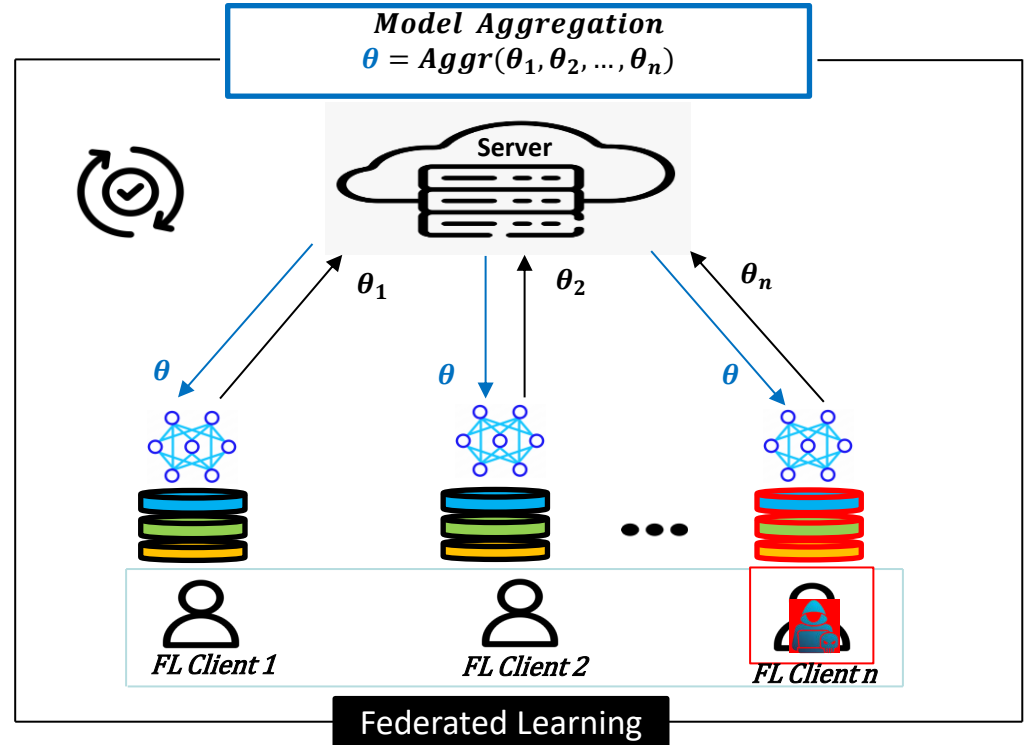
Byzantine attacks

- **Data poisoning** (*manipulate data*)
- **Model poisoning** (*modify gradients*)



Byzantine attacks

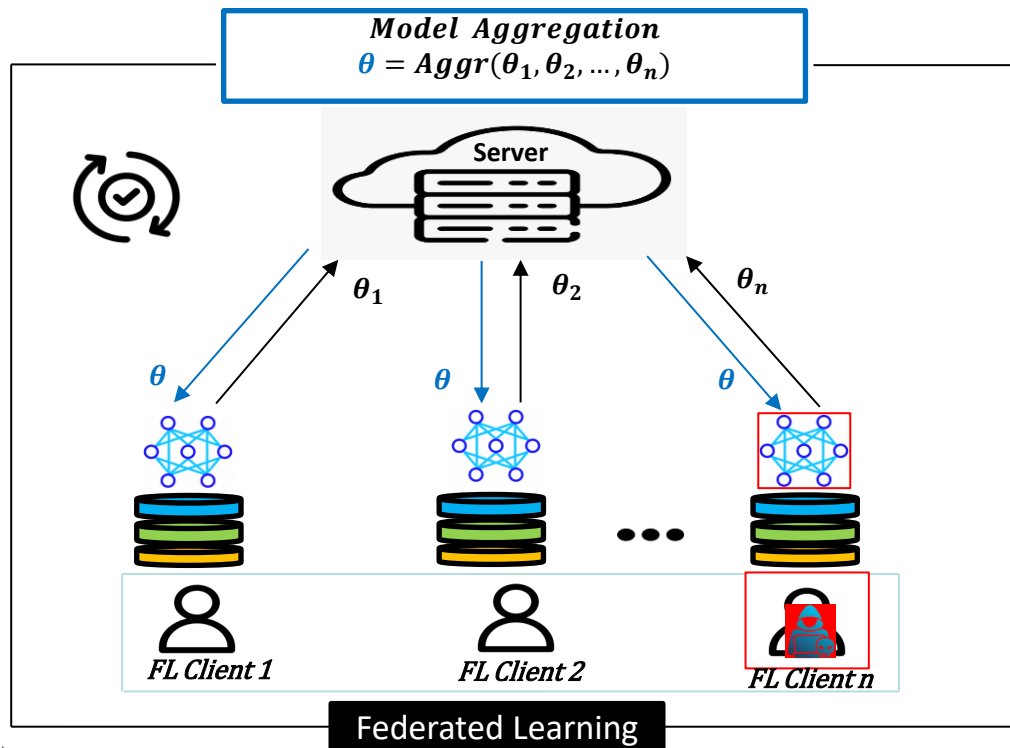
- **Data poisoning** (*manipulate data*)
 - Label Flipping (LF) Attack [Allen-Zhu et al.]
 - Flip dataset labels before training
- **Model poisoning** (*modify gradients*)



Allen-Zhu, Z., Ebrahimi, F., Li, J., and Alistarh, D.(2020). Byzantine-resilient non-convex stochastic gradient descent. arXiv preprint arXiv:2012.14368.

Byzantine attacks

- **Data poisoning** (*manipulate data*)
 - Label Flipping (**LF**) Attack [Allen-Zhu et al.]
 - Flip dataset labels before training
- **Model poisoning** (*modify gradients*)
 - Fall of Empire (**FOE**) attack [Xie et al.]
 - Boost gradients
 - A Little Is Enough (**ALIE**) attack [Baruch et al.]
 - Bias gradients
 - **Mimic** attack [Karimireddy et al.]
 - Mimic honest clients

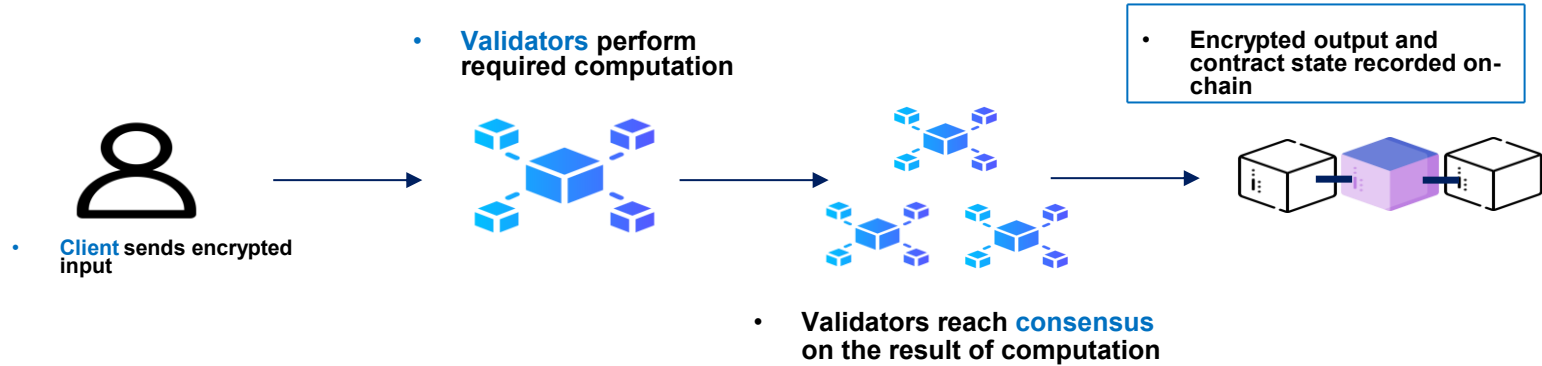


Allen-Zhu, Z., Ebrahimi, F., Li, J., and Alistarh, D. (2020). Byzantine-resilient non-convex stochastic gradient descent. arXiv preprint arXiv:2012.14368.
Xie, C., Koyejo, O., and Gupta, I. (2020). Fall of empires: Breaking byzantine-tolerant sgd by inner product manipulation. In Uncertainty in Artificial Intelligence, pages 261–270.
Baruch, G., Baruch, M., and Goldberg, Y. (2019). A little is enough: Circumventing defenses for distributed learning. Advances in Neural Information Processing Systems, 32.
Karimireddy, S. P., He, L., and Jaggi, M. (2020). Byzantine-robust learning on heterogeneous datasets via bucketing. arXiv preprint arXiv:2006.09365.

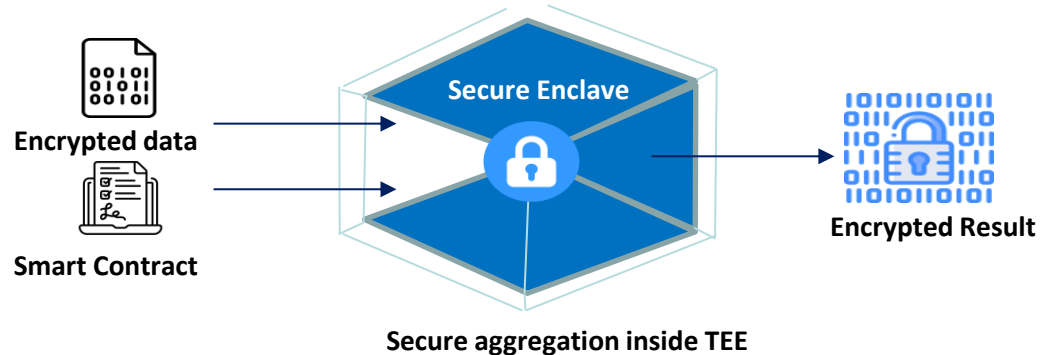
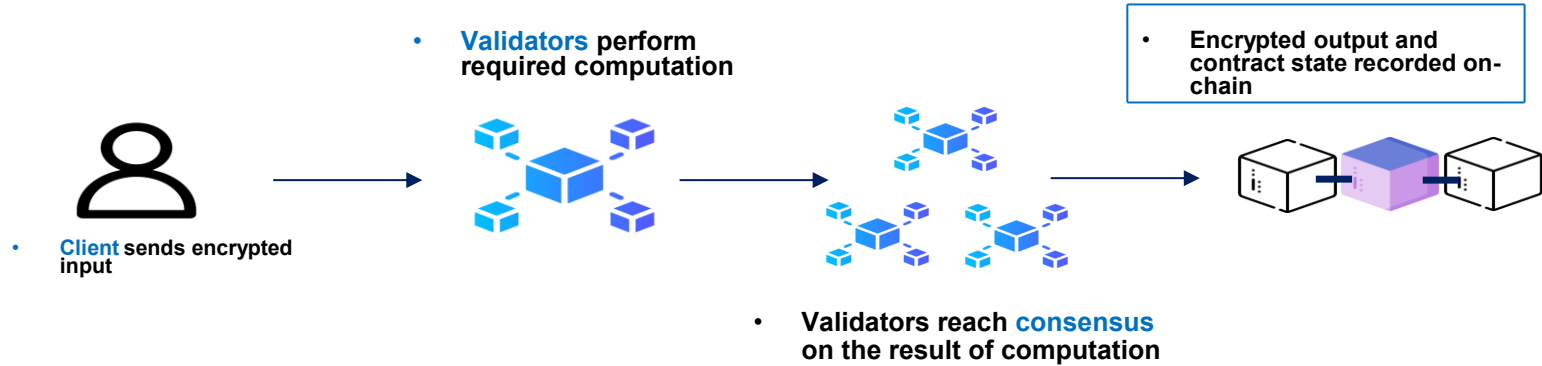
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Blockchain for integrity

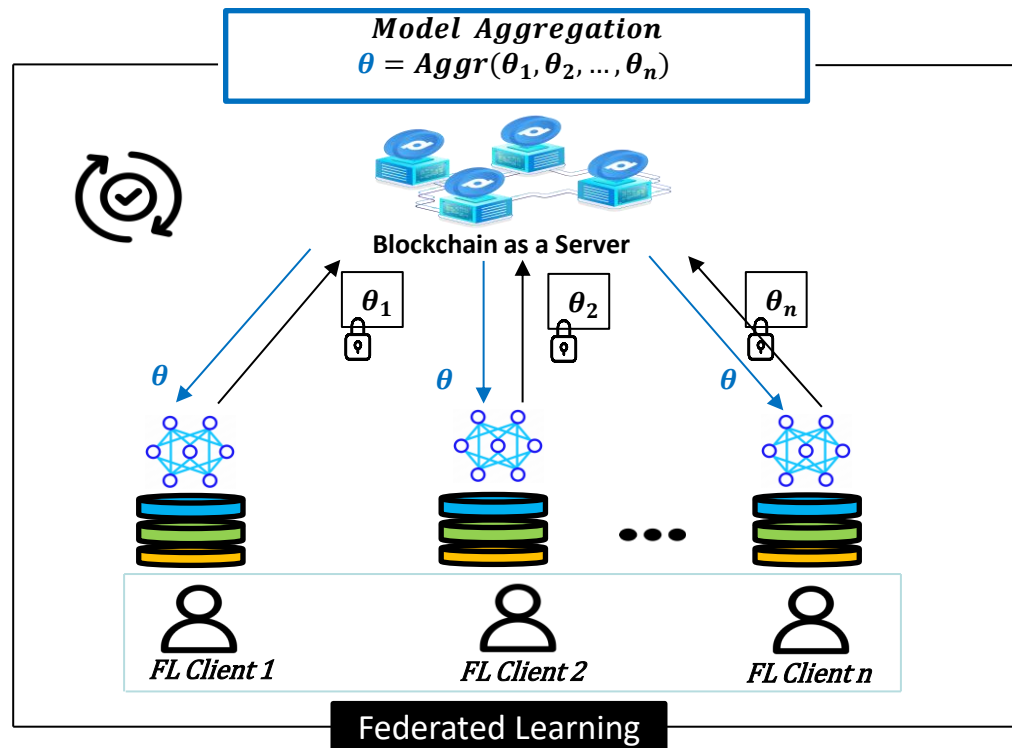


TEE for privacy



Blockchain based solution

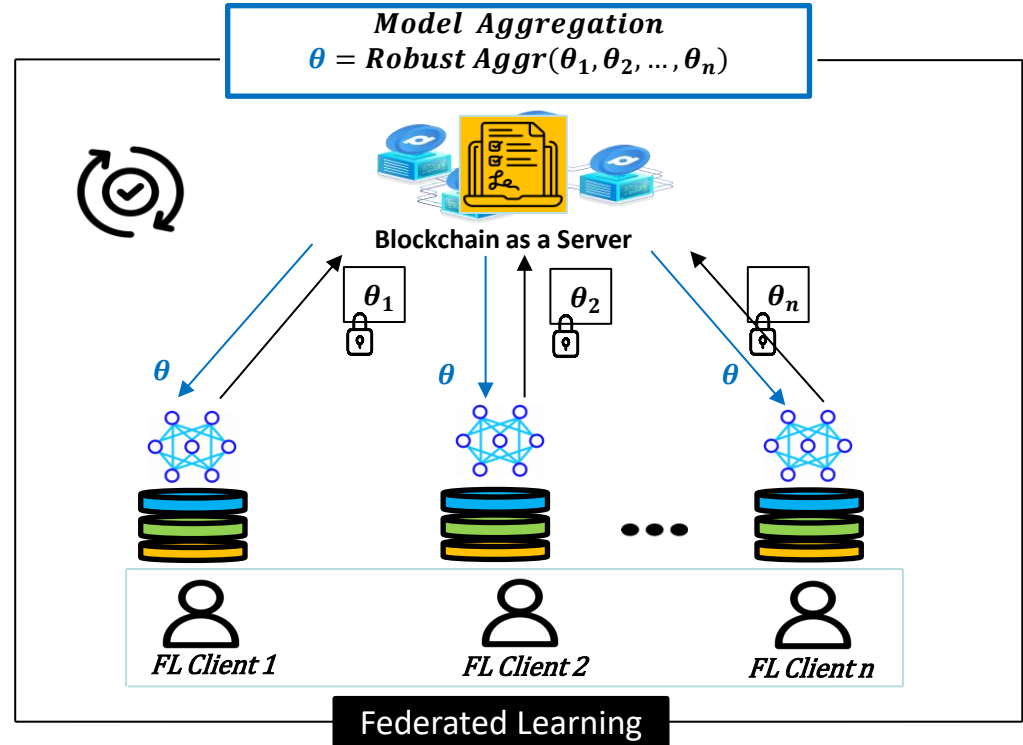
- **Server = Secret Network blockchain**¹
- **Data encryption**
 - Clients encrypt gradients
- **Secure aggregation**
 - Inside TEE



¹<https://scrt.network/>

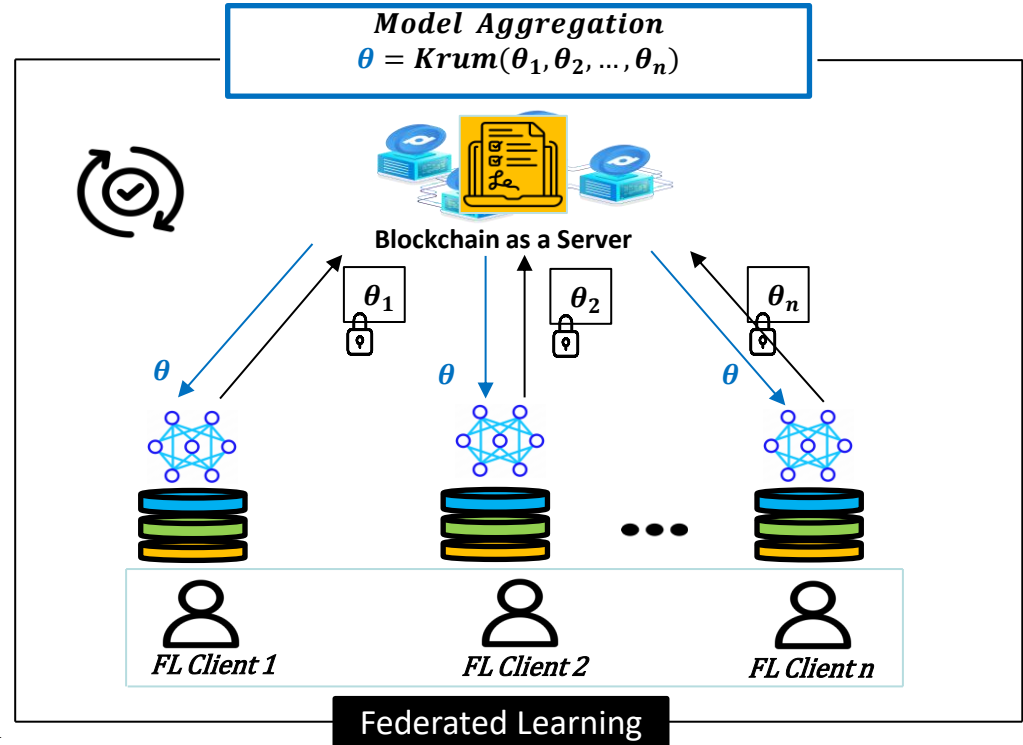
Smart contract for Byzantine-robustness

- **Smart contract**
 - Based on robust aggregation



Smart contract for Byzantine-robustness

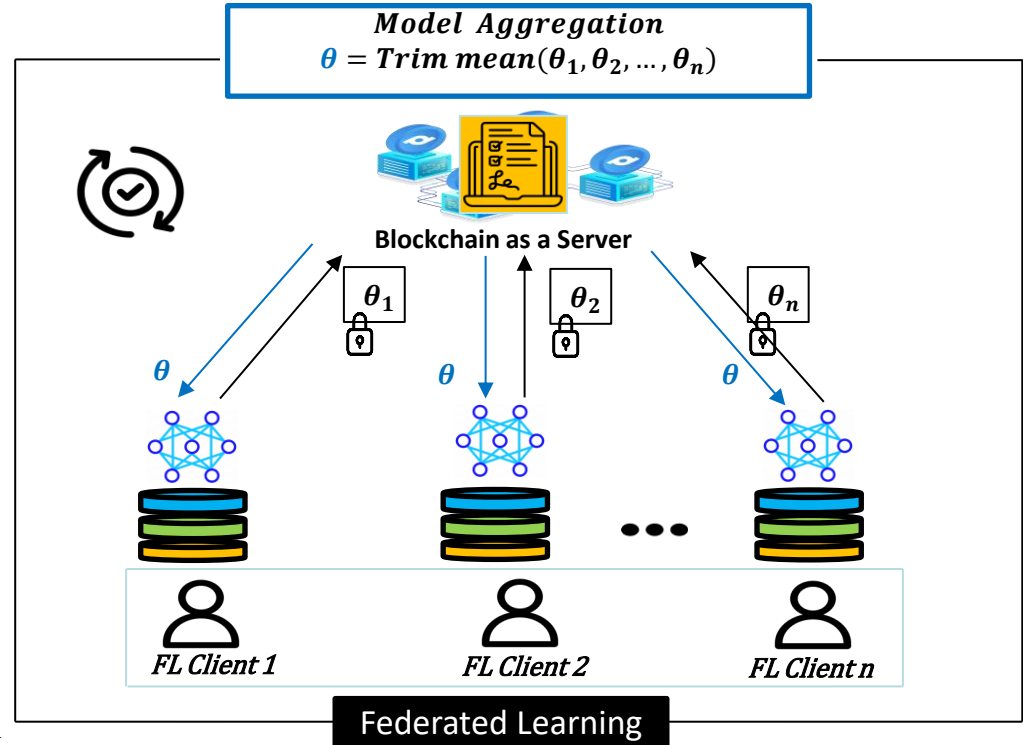
- **Smart contract**
 - Based on robust aggregation
- **Krum** [Blanchard et al.]
 - Compute $\|\theta_i - \theta_j\|^2$
 - Exclude t outliers
 - Global model = Closest clients' gradient



Blanchard, P., El Mhamdi, E. M., Guerraoui, R., and Stainer, J. (2017). Machine Learning with adversaries: Byzantine tolerant gradient descent. Adv. in Neural Information Processing Systems, 30:118–126.

Smart contract for Byzantine-robustness

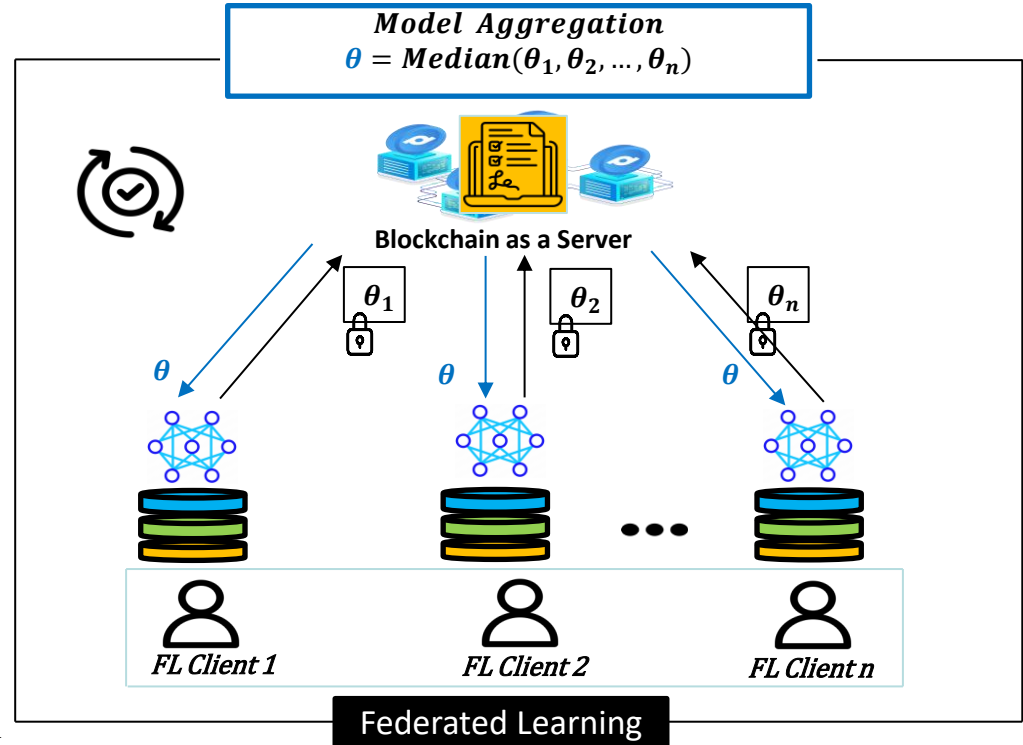
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 - Compute $\|\theta_i - \theta_j\|^2$
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- **Trim mean** [Yin et al.]
 - Sort gradients, remove t extreme outliers
 - Global model = Remaining gradients' mean



Blanchard, P., El Mhamdi, E. M., Guerraoui, R., and Stainer, J. (2017). Machine Learning with adversaries: Byzantine tolerant gradient descent. *Adv. in Neural Information Processing Systems*, 30:118–126.
Yin, D., Chen, Y., Kannan, R., and Bartlett, P. (2018) Byzantine-robust distributed learning: Towards optimal statistical rates. In *Int. Conf. on Machine Learning*, pages 5650–5659.

Smart contract for Byzantine-robustness

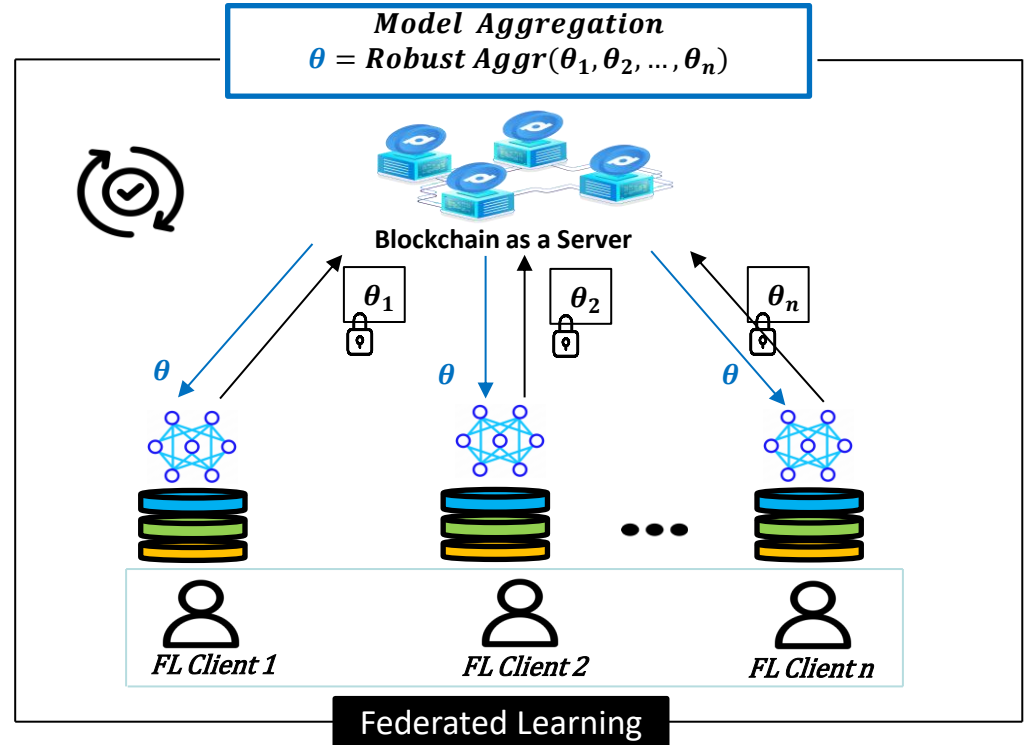
- **Smart contract**
 - Based on robust aggregation
- **Krum** [Blanchard et al.]
 - Compute $\|\theta_i - \theta_j\|^2$
 - Exclude t outliers
 - Global model = Closest clients' gradient
- **Trim mean** [Yin et al.]
 - Sort gradients, remove t extreme outliers
 - Global model = Remaining gradients' mean
- **Median** [Yin et al.]
 - Sort gradients
 - Global model = Median of gradients



Blanchard, P., El Mhamdi, E. M., Guerraoui, R., and Stainer, J. (2017). Machine Learning with adversaries: Byzantine tolerant gradient descent. *Adv. in Neural Information Processing Systems*, 30:118–128.
Yin, D., Chen, Y., Kannan, R., and Bartlett, P. (2018). Byzantine-robust distributed learning: Towards optimal statistical rates. In *Int. Conf. on Machine Learning*, pages 5650–5659.

Summary

- Aggregation Integrity via consensus
- TEE for local model privacy
- Byzantine-robust aggregation



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 - Model Accuracy
 - Overhead Cost

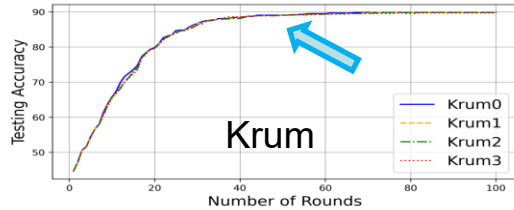
Experimental evaluation

Experimental setup

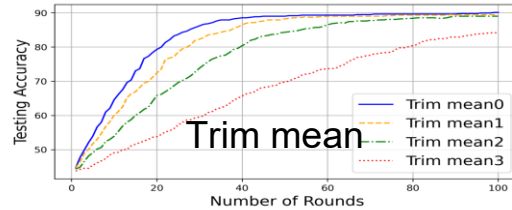
Data Sets	Number of Clients (n)	Byzantine Clients (f)	Attacks
Medical dataset (2126)	10	Up to 30%	FOE Attack ALIE Attack
MNIST (70,000)	15	Up to 30%	LP Attack Mimic Attack

- Model accuracy
- Overhead

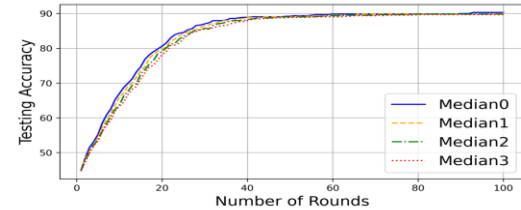
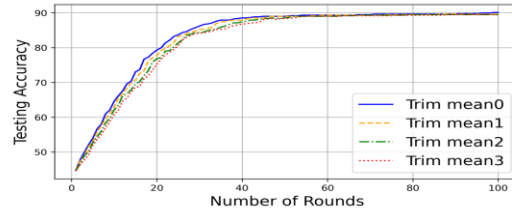
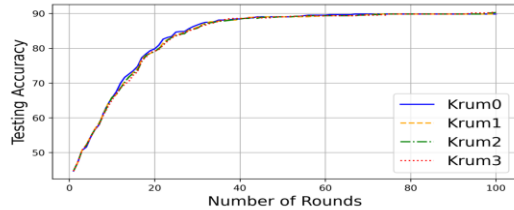
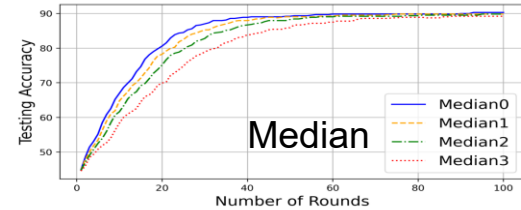
Accuracy over medical dataset



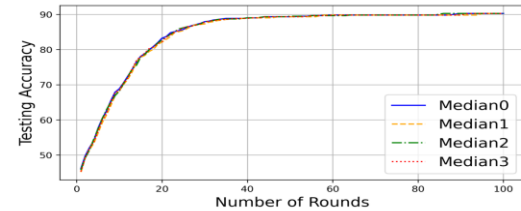
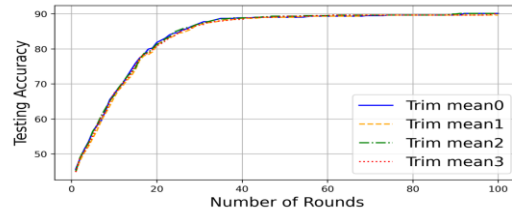
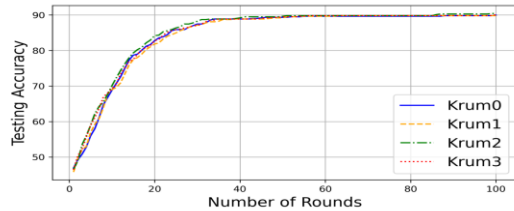
Krum outperforms



FOE Attack

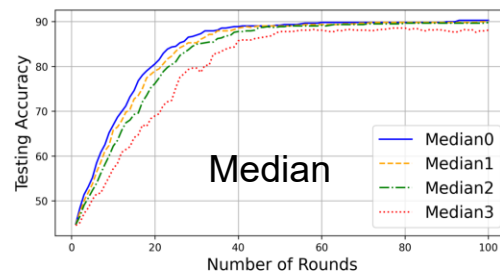
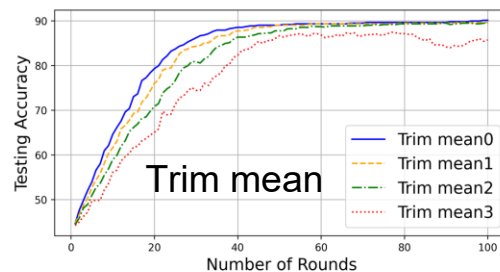
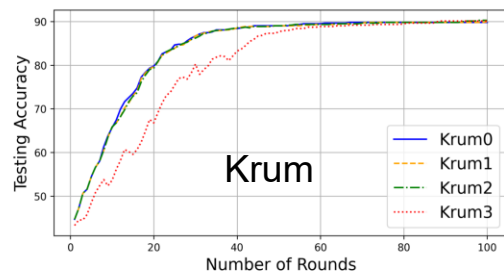


ALIE Attack



LF Attack

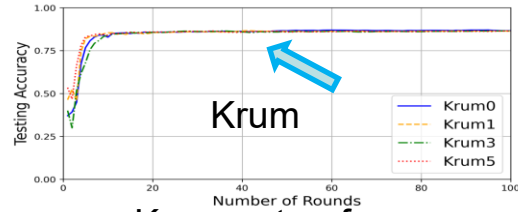
Accuracy over medical dataset



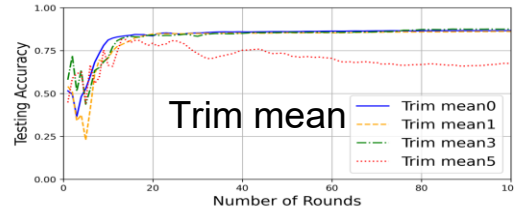
Mimic Attack

Mimic attack is the most severe attack among all

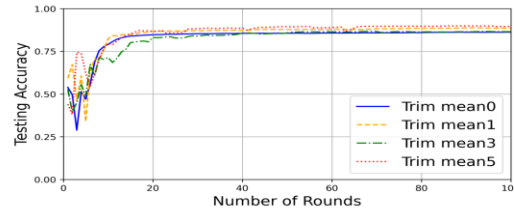
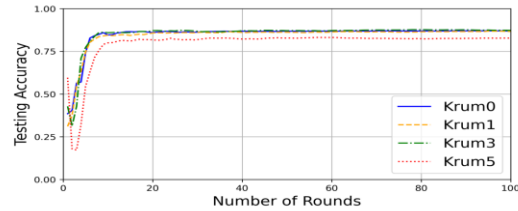
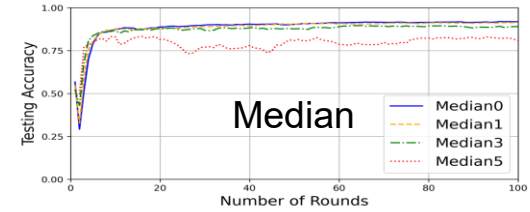
Accuracy over MNIST dataset



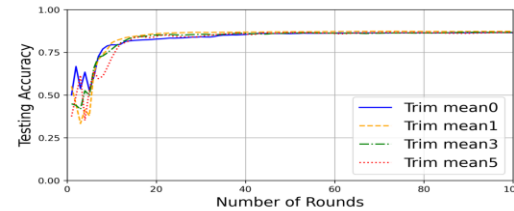
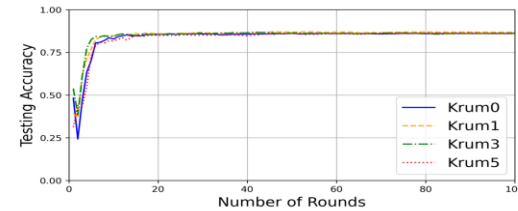
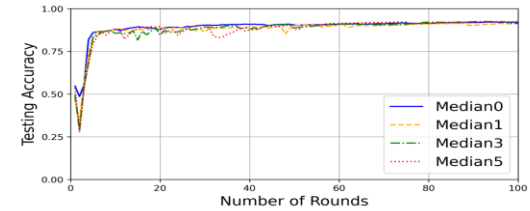
Krum outperforms



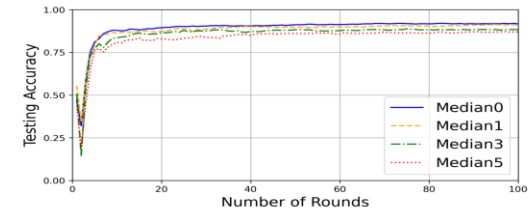
FOE Attack



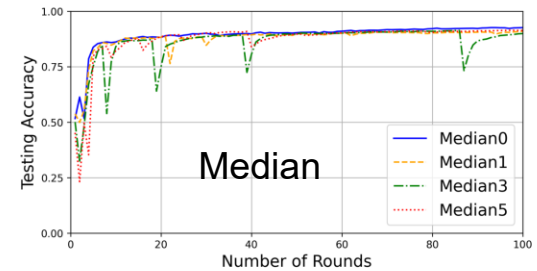
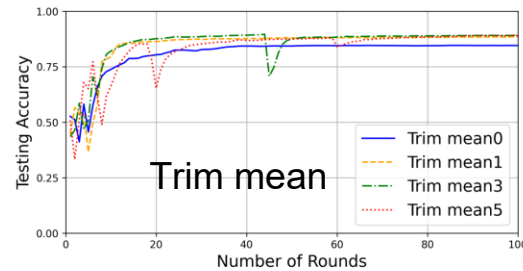
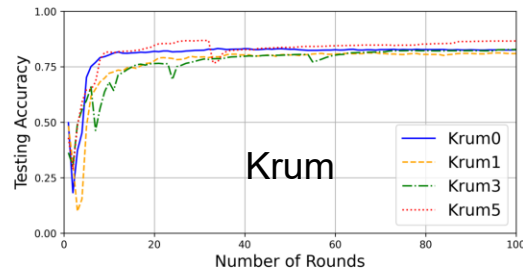
ALIE Attack



LF Attack



Accuracy over MNIST dataset



Mimic Attack

Mimic attack is the most severe attack among all

Medical dataset overhead

Techniques	t_{aggr} (s)	BW (KB)
Krum		
Trim mean		
Median		

Medical dataset overhead

Techniques	t_{aggr} (s)	BW (KB)
Krum	4.32	
Trim mean	4.11	
Median	4.20	

Medical dataset overhead

Techniques	t_{aggr} (s)	BW (KB)
Krum	4.32	13.67
Trim mean	4.11	13.67
Median	4.20	13.67

Medical dataset overhead

System	Techniques	t_{aggr} (s)	BW (KB)
Our Work	Krum		
	Trim mean		
	Median		

MNIST dataset overhead

System	Techniques	t_{aggr} (s)	BW (KB)
Our Work	Krum	7.68	
	Trim mean	7.06	
	Median	6.71	

MNIST dataset overhead

System	Techniques	t_{aggr} (s)	BW (KB)
Our Work	Krum	7.68	19.42
	Trim mean	7.06	19.42
	Median	6.71	19.42

MNIST dataset overhead

System	Techniques	t_{aggr} (s)	BW (KB)
Our Work	Krum	7.68	19.42
	Trim mean	7.06	19.42
	Median	6.71	19.42
SABLE [Choffrut et al.]	Trim mean	1040.40	5

Choffrut, A., Guerraoui, R., Pinot, R., Sirdey, R., Stephan, J., and Zuber, M. (2023). Practical homomorphic aggregation for Byzantine ML. arXiv preprint arXiv:2309.05395.

Conclusion & Future Work

- **Conclusion**

- Aggregation integrity, privacy and Byzantine attacks in FL
 - Integrity via blockchain
 - TEE for privacy
 - Byzantine-robustness via smart contract
- Mimic attack is the most severe attack
- Krum outperforms

- **Future Work**

- Integrating other robust mechanisms
 - L_2, L_∞ , Cosine distance

Thank you

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